Mohammad Ninad Mahmud Nobo

Bangladesh University of Engineering and Technology (BUET) | CSE

Email: mninadmnobo@gmail.com GitHub: github.com/mninadmnobo LinkedIn: linkedin.com/in/mninadmnobo
Portfolio: mninadmnobo.github.io Scholar: Google Scholar (M Ninad M Nobo) Phone: +880 1939-444451

Summary

Computer Science graduate specialized in Artificial Intelligence, Machine Learning, Large Language Models, and software engineering across research and industry. Skilled in building production-grade software systems, including deep learning models, multi-agent AI pipelines, and web applications. Research spans LLM-based software testing, medical AI and applied machine learning.

Education

Bangladesh University of Engineering and Technology (BUET) **2022 – 2026**
Bachelor of Science in Computer Science and Engineering (CSE) **CGPA: 3.60 / 4.00**

Relevant Coursework: Data Structures and Algorithms, Algorithm Engineering, Database Systems, Operating Systems, Computer Architecture, Computer Networks, Compiler Design, Artificial Intelligence, Machine Learning, Computer Graphics

Research Interests: Large Language Model Systems, AI-Assisted Software Engineering, Applied Machine Learning, Medical AI

Professional Experience

- **AI Software Engineer**
  Department of Research and Development (R&D) - Saturn Textiles Limited *2026 – Present*

  Web Development [Saturn R&D Website | GitHub] [FABINS Product Portfolio | GitHub]
  Next.js • React • Spring Boot • REST APIs • Brevo • GitHub Actions *2026*
  – Developed Saturn R&D Website and FABINS Product Portfolio with integrated form submission and email notifications.

  FABINS - Fabric Inspection Automation | ML & Computer Vision
  Python • PyTorch • Computer Vision • Deep Learning *2026 – Present*
  – Developing computer vision and deep learning models for automated fabric defect detection and quality inspection.

Research Experience

- **AutoTestGenX: Multi-Agent LLM Framework for End-to-End Web Testing (Undergraduate Thesis)** [GitHub]
  Python • LLM Engineering • Multi-Agent Systems • Web Testing • Browser Automation *2025 – 2026*
  – Designed a multi-agent LLM framework to automate web test generation and execution from natural-language requirements.
  – Built benchmark datasets and evaluated against ground-truth suites, achieving 84.0% scenario coverage and 90% error detection.
- **MedCAR: Conflict-Aware Medical Reasoning System** [GitHub]
  Python • Deep Learning • Medical AI *2026*
  – Built a multi-model chest X-ray reasoning system with a conflict-resolution pipeline to reconcile contradictory model outputs.
  – Designed a framework using confidence calibration and uncertainty-based abstention for reliable predictions.

Academic Projects

- **MindTrace: AI-Powered Dementia Care Application** [GitHub | Feature | Infrastructure]
  Kotlin • Spring Boot • Spring AI • PostgreSQL • Redis • Firebase • Docker • Azure *2025*
  – Developed an Android app for dementia patients and caregivers with personalized assistance and caregiver support features.
  – Built the Spring Boot backend and designed an accessible Android UI in Kotlin for smooth and intuitive AI interactions.
- **Gemma VetCare: AI-Assisted Farming Support System** [GitHub | Feature]
  Kotlin • Android Studio • Spring Boot • Spring AI • MongoDB *2025*
  – Built an AI-assisted Android application delivering real-time veterinary diagnosis and treatment recommendations.
  – Designed RESTful APIs optimized for low-connectivity environments with efficient request handling.
- **TCP Window Scaling Attack** [GitHub]
  Python • TCP/IP • Network Security • Wireshark • Linux *2025*
  – Built a virtualized MITM lab to demonstrate ARP poisoning and TCP window scaling attacks between client-server VMs.
  – Implemented attack and defense clients and analyzed TCP traffic using Wireshark and packet captures.
- **Smart Gardening Automation** [GitHub | Feature]
  Arduino • Embedded C++ • Sensors • Bluetooth Communication • Automation *2024*
  – Developed a gardening automation system integrating environmental sensing and wireless communication.
  – Implemented automated watering, fertilizer/pesticide cycles, smart shade control, and remote monitoring.

Technical Skills

AI/ML: PyTorch, TensorFlow, Scikit-learn, Large Language Models, Prompt Engineering, Multi-Agent Systems, Computer Vision
Backend & Full-Stack: Spring Boot, Spring AI, Next.js, React, Node.js, Express.js, REST APIs, Android
DevOps & Tools: Linux, Git, Docker, Docker Compose, GitHub Actions, Firebase, Azure
Languages: Python, C/C++, Java, JavaScript, Kotlin, SQL
Databases: PostgreSQL, MongoDB, Redis

Additional Information

Spoken Languages: Bangla (Native), English (Fluent), Hindi & Urdu (Conversational), Arabic (Reading)
Programming Profiles: NeetCode: mninadmnobo LeetCode: mninadmnobo
Research Profiles: ORCID: 0009-0006-2781-6693 ResearchGate: Mohammad Ninad Mahmud Nobo